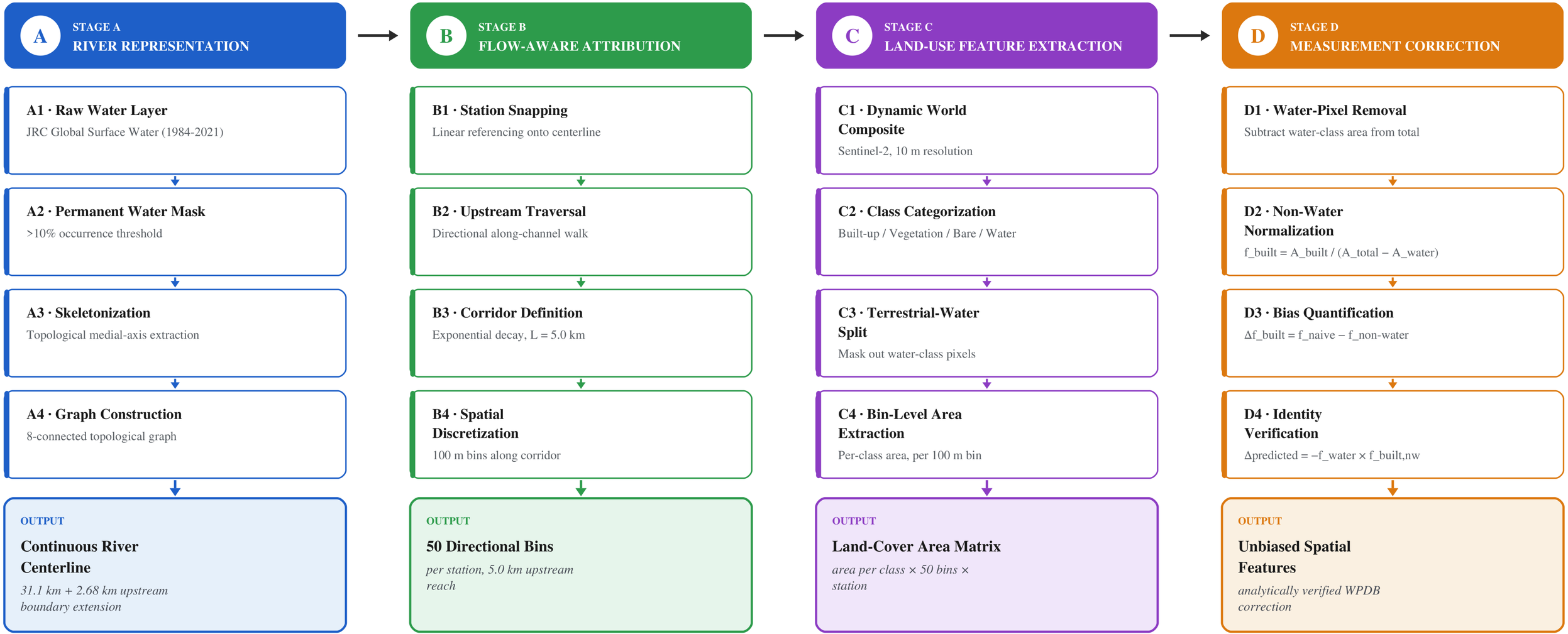

A
STAGE A
RIVER REPRESENTATION
A1 · Raw Water Layer
JRC Global Surface Water (1984-2021)
A2 · Permanent Water Mask
>10% occurrence threshold
A3 · Skeletonization
Topological medial-axis extraction
A4 · Graph Construction
8-connected topological graph
OUTPUT
Continuous River Centerline
31.1 km + 2.68 km upstream boundary extension
B
STAGE B
FLOW-AWARE ATTRIBUTION
B1 · Station Snapping
Linear referencing onto centerline
B2 · Upstream Traversal
Directional along-channel walk
B3 · Corridor Definition
Exponential decay, L = 5.0 km
B4 · Spatial Discretization
100 m bins along corridor
OUTPUT
50 Directional Bins
per station, 5.0 km upstream reach
C
STAGE C
LAND-USE FEATURE EXTRACTION
C1 · Dynamic World Composite
Sentinel-2, 10 m resolution
C2 · Class Categorization
Built-up / Vegetation / Bare / Water
C3 · Terrestrial-Water Split
Mask out water-class pixels
C4 · Bin-Level Area Extraction
Per-class area, per 100 m bin
OUTPUT
Land-Cover Area Matrix
area per class × 50 bins × station
D
STAGE D
MEASUREMENT CORRECTION
D1 · Water-Pixel Removal
Subtract water-class area from total
D2 · Non-Water Normalization
f_built = A_built / (A_total − A_water)
D3 · Bias Quantification
Δf_built = f_naive − f_non-water
D4 · Identity Verification
Δpredicted = −f_water × f_built,nw
OUTPUT
Unbiased Spatial Features
analytically verified WPDB correction